QUYNH VO

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Education

Carleton University *Ottawa, ON*
Bachelor of Computer Science, Stream in Software Engineering (with High Distinction) *GPA: 3.9*

Experience

Full Stack Software Developer Co-op | **Ciena** *Ottawa, ON | Jan 2026 – May 2026*

- Engineered a **Python CLI** managing manifests across **60+ microservices**, automating data processing by resolving JSON inheritance graphs with **cycle detection** to surface configuration drift via HTML reports.
- Automated provisioning across **10 distributed products** and **15+ parameters** by building a bidirectional CLI serialization engine (**React, Redux, Python**) integrated with the **Bitbucket API**
- Improved system reliability across distributed service domains by building real-time observability dashboards (**WebSockets, Redux**) for memory profiling and heap diagnostics, working on **Linux** machines via **SSH** and **TigerVNC**.

Software Developer Co-op | **Ericsson** *Ottawa, ON | May 2023 – Aug 2024*

- Improved UI accessibility across a network planning tool by building a vector-based floorplan editing feature **(GoJS, React, Redux)**, and introduced **TypeScript** to cut runtime errors by 25%.
- Integrated **REST APIs** for real-time network configuration and investigated **Azure AD authentication** and **identity flows** for a scalable login solution while resolving production UI defects.

Software Developer Co-op | **Ciena** *Ottawa, ON | Sep 2022 – Dec 2022*

- Improved dashboard load performance by **15%** by building scalable UI components and RESTful API interfaces for MCP network applications in **TypeScript and Ember.js.**
- **Automated test suites and CI/CD workflows**, accelerating deployment cycles and improving release reliability across distributed network and services.

Projects

Museboard | *C++, OpenCV, ONNX Runtime, FastAPI, PostgreSQL/pgvector, React* *Aug 2026*

- Built a mood-board website that **finds similar images by visual content or color** using **C++, React, FastAPI, PostgreSQL/pgvector,** containerized with **Docker Compose**. Designed a **background processing pipeline** with **2 C++ CPU workers** and **1 AI worker** to generate thumbnails, color palettes, and CLIP embeddings using **OpenCV and ONNX Runtime**, backed by a persistent job queue with automatic retries and stale-result rejection.

Transit Routing & Compression Engine | *C++, CMake, Dijkstra, GTFS, Data Compression* *June 2026*

- Developed a **C++ transit trip-planning engine** over **Montreal's STM network** (9,188 stops, millions of GTFS records) with **delta/varint compression** achieving 79.5% coordinate size reduction, sub-meter accuracy.

AI Test Coverage Analyzer (Ciena Hackathon) | *Python, Google ADK, Gemini 2.5, Pydantic* *April 2026*

- Built a **multi-agent AI backend pipeline** (**Google ADK, Gemini 2.5 Flash, asyncio**) combining deterministic Pandas ingestion with parallel LLM calls and **Pydantic** schema enforcement to audit test coverage across **500+ files**, reducing manual review by **4–6 hrs/release**.

Smart Appetite Manager (1st Place uOttaHack8) | *Python, Agentic AI, Solace Agent Mesh* *Jan 2026*

- Built an **event-driven agentic AI platform** (Python, React, Solace Agent Mesh), with 4 orchestrators & 9 specialized agents to track inventory, recommend recipes, and compare prices across 13 grocery chains in a **36-hour sprint.**
- Built a **multi-step LLM enrichment pipeline** combining text filtering, categorization, and vision-model extraction to **reduce noisy API** results from **77 to 15** relevant products per query.

Technical Skills

Languages: Python, JavaScript/TypeScript, C++, Java, SQL, HTML/CSS

Frameworks & Libraries: React, Redux, Next.js, Node.js, Flask, Material-UI, Ember.js, Pandas, Google ADK

Developer Tools: Git, CI/CD (GitLab CI), RabbitMQ, WebSocket, REST APIs, Linux, Valgrind, GDB, Jenkins

Cloud & Platforms: Google Cloud Platform (GCP), Bitbucket Server, Solace PubSub+, AI/LLM APIs